

Ivan Petrov

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SUMMARY

Go engineer with a background in systems and C++ embedded/robotics. Built production gRPC/protobuf services, REST backends, and TUI tooling

WORK EXPERIENCE

Software Engineer

Northwind Robotics | Jan. 2025 - Present | *Berlin, Germany*

- Built a Go service that generates **protobuf** schemas from C++ headers at build time and bridges **gRPC** <-> a legacy binary telemetry protocol: sustains **12k RPC/s** on a single ARM board across a fleet of vehicles
- Refactored and extended a 2M+ LOC **Qt/QML** ground control station for robotic systems: custom QML screens and C++ integrations in a 7-person team
- Designed and implemented an internal **artifact storage service** in Go: REST API, Redis for metadata, Docker volume as file store, consumed by a React frontend - replaced manual binary distribution
- Developed a USB-over-IP relay on a single-board **ARM Linux** computer for the radio subsystem, achieving **80 MB/s** sustained throughput

Software Engineer (Contract)

Contoso Avionics | Sep. 2024 - Jan. 2025 | *Remote*

- Designed system architecture and developed the **Flutter** operator interface with a custom **WebSocket** protocol for real-time ROS integration

CAE / Software Engineer

Fabrikam Propulsion | Mar. 2023 - Aug. 2024 | *Munich, Germany*

- Developed **C++** firmware for an **STM32F407**-based engine test stand: sensor data acquisition, SD logging, and UART telemetry output

PROJECTS

nightglass

Go, GStreamer, ARM Linux, NPU

- Building a DIY digital night-vision helmet on a **6 TOPS NPU** board: dual thermal cameras fused with a near-infrared stream, on-device **YOLOv8n** person detection - targeting **<100 ms** end-to-end latency, primary language Go

pgscope

Go, TUI, PostgreSQL | github.com/yourhandle/pgscope

- Keyboard-driven read-only TUI for exploring PostgreSQL, designed to run on the server and accessed over SSH - no port forwarding or local client setup
- Component-based **Bubble Tea** architecture: schema browser and live table data viewer
- All queries run in **REPEATABLE READ** read-only transactions - consistent snapshot with no risk of accidental mutation

vpn-ansible

Ansible, DevOps | github.com/yourhandle/vpn-ansible

- Ansible playbook for automated **WireGuard** VPN provisioning in Docker: idempotent per-client key and config generation, stable IPs across re-deploys
- Multi-distro support (Debian, RedHat, Arch, Alpine) - from bare VPS to running VPN in a single command

TECHNICAL SKILLS

Languages: Go, C/C++, Python, SQL

Technologies: Qt/QML, WebSocket, Ansible

Databases: PostgreSQL, Redis, SQLite

Developer Tools: Git, GitHub Actions, GitLab CI, Docker / Podman, Linux

EDUCATION

Example State University

M.Sc., Computational Mathematics and Computer Science | Aug. 2020 - Jun. 2022 | *Berlin, Germany*

Institute of Applied Technology

B.Sc., Geoinformatics, Expected 2029 | Aug. 2022 - Jan. 2029 | *Munich, Germany*